FULL-SYSTEM-MASTER-PLAN — one document, the whole machine (Session 45, 2026-10-29)

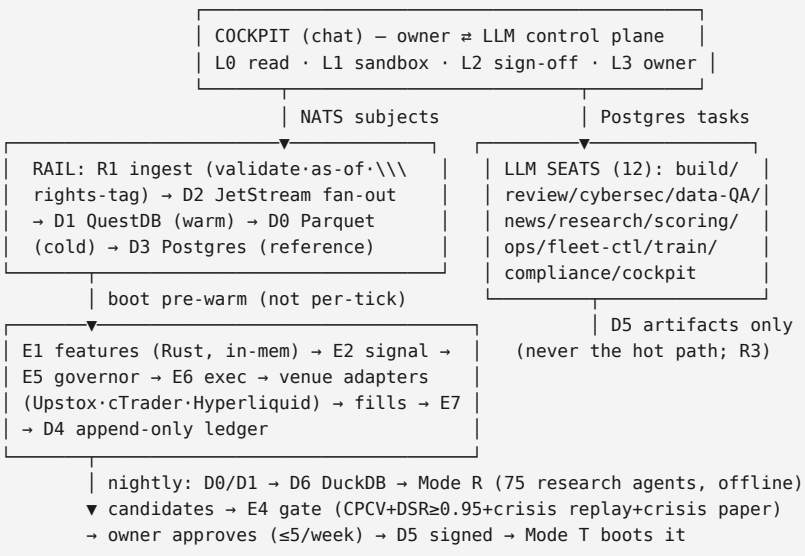
> Status: **PLAN** — the single entry-point that unifies the five specialist docs into one build plan: structure, architecture, workflows, memory, comms, quality gates, and the stage-by-stage build instructions. Nothing here relaxes any existing decision; it consolidates them and adds the connective tissue that was missing (end-to-end workflows, the agent-teams operating model, monitor/control chat, and per-stage validation gates). Labels: **ESTIMATE** = projected, to be measured; **MEASURED** = command output on record; everything else traces a source doc.

§0 · Reading map — where each decision lives

Topic	Source of truth	This doc's role
Whole-ecosystem rows, R1–R8 rules, 10-stage order	`FULL-SYSTEM-PLAN.md`	restated §1/§2
The 75 non-LLM trading cells, guards, runbooks, training	`AGENT-FLEET-BLUEPRINT.md`	restated §3/§6/§7-lite
Store ladder D0–D7, engines E1–E7, wiring, Nautilus verdicts	`ENGINE-DATABASE-PLAN.md` (+§19 TSDB final)	restated §4/§5
LLM control-plane (12 seats, orchestration, scoring gate)	`LLM-FLEET-BLUEPRINT.md`	restated §3/§7-lite
Dataset gathering (1,114-row master TODO)	`Research/data/todo/` master + §17/§18 of engine plan	consumed §4-B1
Everything together — workflows, teams, memory, comms, gates	**this document**	the missing "how it all runs and builds" layer

Committed sessions live in `planing conva/PLANNING-CONVERSATION-LOG.md` (S1→S45); open decisions O3–O16 are reproduced in §14 with owners/waves.

§1 · The one-page system picture



Three rows still hold (FULL-SYSTEM-PLAN §2): Row 1 = Terminal + News portal + Media hub (public, rights-firewalled, LAST); Row 2 = the three engines (Financial · News · World) on the rail above; Row 3 = FinOS (this trading system) + Cockpit + World Monitor (last of all).

§2 · The two non-negotiable rule families

Architecture rules R1–R8 (FULL-SYSTEM-PLAN §3): collect once/reuse many · rights firewall at Row 1 boundary only · **intelligence never in fast paths** · PIT + survivorship discipline · provenance everywhere · versioned contracts · deployment gates (train→eval→paper→approve→deploy) · HFT honesty (venue RTT dominates; internal budget <5 ms; retail feeds are research-grade).

Fleet guardrails (AGENT-FLEET-BLUEPRINT): guards G1–G6 — G1 cells cannot edit their own artifacts · G2 no search code in Mode T · G3 research can't place orders · G4 every trial logged · G5 one active style per symbol · G6 Mode R/T separate processes — plus the L0–L4 kill cascade (cell→symbol→team→market→global) and an out-of-band watchdog kill that works without the app · startup invariant: reconcile-before-orders · idempotent order IDs · D4/D5 append-only/signed at the DB-role level.

Regulatory R9–R14 and the venue-specific non-negotiables stay as in AGENT-FLEET-BLUEPRINT §7 (Upstox algo-tagging is O9; Hyperliquid FIU-IND/FEMA is O11; ToS/ban checklist is O13).

§3 · The workforces — "one team of agents"

3.A The non-LLM fleet: 75 brains, ~480–545 hands, ≤75 active

Market	Symbols	Hands (cells ceiling)	Exits
Stocks (NSE/BSE, options buying only)	≤59	~440–500	rule ladder 1–6 + kill
FX Team 1 (cTrader ECN)	11	29	cascade-flatten, funding-window print, kill
FX Team 2 (separate account, pure scalping)	2	2	push-mode, session-closed flat
Crypto (Hyperliquid perps)	3	10–12	funding/liquidation/weekend/PAIR exits

*Owner says "73" — that was the old symbols×1-style count (FLEET §1 note). The locked roster is **75 research brains**; the hands (cells) get built only when E4 promotes a style, and G5 keeps exactly one alive per symbol.*

3.B The LLM control plane: 12 seats (all R3; never the hot path)

Row extracted verbatim in `LLM-FLEET-BLUEPRINT.md` §2: build · review · cybersec · data-QA · news · research/analysis · scoring · ops · fleet-control · training · compliance · cockpit/comms. Hard ceilings: seats produce **PRs/findings/artifacts/asks for approval** — never market writes outside L0–L3, never frozen-artifact edits without E4, never an order from research.

3.C How the three work together (the operating loop)

Loop	Trigger	LLM seat	Fleet side	Output
Build	owner or W-wave task	build seat writes code+tests	never runs live	PR → review seat → cybersec seat → owner merge
Bug/issue fix	alert or owner chat	build seat reproduces (flight-recorder replay)	cell frozen for the fix window	fix PR + regression test
Score/promote	scheduled research	scoring seat prepares D5 candidate	E4/E2/E5 run the gates	owner approval (≤5/week)
Research	nightly or on ask	research seat (citations required)	E1/E4 replay	cited artifact in D5
News/event	market hours	news seat digests	news ladder + kill-floor checks R3 records	digest + R3 record
Data-QA	nightly + rail alerts	data-QA seat	gap calendar	tickets, never silent fixes
Ops/kill	any L0–L4 trigger	ops seat narrates	E5/E7 run it	incident report + runbook update
Train	weekly + drift	training seat	G4/D5 machinery	candidate artifacts through E4
Monitor	cockpit session	fleet-control seat	NATS state map	chat + FL0–L4 queue
Self-learn	weekly	training + research	replay metrics	proposal set only

The honest division: LLM seats *decide-and-explain*, the deterministic fleet *executes-and-proves*. Nothing an LLM seat says becomes live unless a gate (E4/E2, DSR, edge-lift, L0–L3) says so.

§4 · What the 13 owner-named repos contribute (deep-read verdicts — measured 2026-10-29)

Per-repo rows with file-level cites live in `LLM-FLEET-BLUEPRINT.md` §3.1; summary:

Contributes to	Repo	Mechanism	Constraints honored
Boss-loop contract	ruflo (`@claude-flow/swarm`)	DAG orchestrator, retries, cycle-detection	in-proc bus = we keep NATS+Postgres durable queue ourselves
Loop discipline	brigade	loop-guards: budget/no-progress/repetition/**independent done-checks**/completion veto	brand-scrubbed fork → vendor-pinned; gateway-runtime decision = open O16
Approval contract	brigade `approval-bridge`	request/resolve tokens for L0–L3	mechanism = ours
Scheduled seats	brigade cron/isolated-agent + metering	headless scheduling pattern	—
Role library	agency-agents (267 personas, already wired in this workspace)	prompt-only, runbooks.json roster contract	sanitize marketing prose
Decision procedure	day-trader-skill	11 SKILL.md + deterministic timemachine/risk/size (315/315 tests pass)	US/long-only scope: merge our instruments (S44 note)
Typed schemas	TradingAgents `agents/schemas.py`	Pydantic signed decision outputs	strip vendors/runtime
Prompt packs	Vibe-Trading `swarm/presets/*.yaml`	hard anti-fabrication data-citation rules	prose not typed → local schema pass
Research grounding	quant-mind	cited-artifact schema (chunk+page+quote+`as_of`); BM25/agentic retrieval	processor, not corpus
Forecast gate	timecopilot	forced beat-SeasonalNaive via output-validator; FunctionModel stub testing = our eval-harness template	Pre-Alpha; don't wholesale-install
Streaming layer candidate	proton	**native NATS JetStream external streams** + streaming materialized views	trial at W2, NOT a D1 change (QuestDB stays D1 per §19)
Cockpit UI ideas	OpenStock	session-clock, alert schema, cache-pool	**AGPL — ideas only, no code lift**
Decision-head contract	laya-mlx	choice/score/noul + abstention + calibration clamps	**runtime REJECT** (MLX darwin/arm64 only, CUDA unsupported in code)
REJECTS	AI-Trader (no LICENSE file — README badge links to nothing; platform, not library) · nixtla (cloud-locked `api.nixtla.io` SDK; free models live in `statsforecast`/`mlforecast`)	—	license/hardware/cloud
Memory recall	Graphify (owner addition)	local AST graph, 0-token code mode, opencode skill install	Cloud upsell rejected; "71.5×" claim = author-reported (§9)

§5 · The data plane (consolidated, decision-final)

Store	Engine	Holds	Decided
D0	Parquet on NVMe + object backup	years of ticks/bars/filings/news/world captures	§18 16-TB target, ESTIMATE-range 4–16 TB
D1	**QuestDB** (final §19; ClickHouse fallback if W0a gate fails twice)	live window only	Timescale rejected for D1 (tsl/ license wall)
D2	NATS JetStream	live event fan-out	proton as layered streaming-MV candidate at W2
D3	PostgreSQL	registry/reference/rights/calendars/limits	+ LLM `llm_tasks` /cockpit tables
D4	PostgreSQL append-only role	fills/P&L/trials/audit ≥5 yr	E7 + G4 write here
D5	filesystem + Postgres index	frozen artifacts, feature-hashed, signed	scoring outputs versioned here
D6	DuckDB over D0/D1	research scans	Mode R only
D7	(upsert) ClickHouse candidate if research outgrows D6	future	not planned now

Gathering: master TODO 1,114 rows (18 volumes; S43 merge) drills wave-tagged W0/W1/ W2/W3; B1 backfill gate = all 3 domains loaded 6–10y + gaps table verified **before any agent trains** (S42/§17.3). Cross-check rows each wave; no walled-source scraping (DB-GATE per §17.2).

§6 · The engine stack (who builds, who adopts)

Role	Engine	Built by	Note
Ingest rail	R1	ours (Rust)	validate·as-of·rights-tag·fan-out writes
Feature engine	E1	**ours (Rust)**, tree-sitter-style determinism + ONNX where ML	pre-warm at boot; no per-tick store query
Signal	E2	ours (Rust)	loads frozen D5 artifacts at boot
Gate	E4	ours (research tooling + CPCV/DSR crates)	council-fixed stats
Governor	E5	ours (Rust)	deterministic
Execution	E6 adapters	**Nautilus** runtime for cell hosts (fleet), adapters ours	Nautilus held the speed verdict (§13)
Ledger	E7	ours	append-only D4 writer
LLM runtime	(seats)	Claude Code / opencode harnesses + brigade-loop/ruflo-swarm ports	never in hot path

Rust↔Python boundary: kernel = Rust (rail/hot path); edge = Python (research, E4 tooling, LLM seats) — monorepo, hard boundary, contracts across (FULL-SYSTEM-PLAN §6).

§7 · Workflows (the "how it all runs," pinned)

7-A One data point's life (session-locked)

venue → R1 (validate·as-of·rights) → D2 (µs–ms fan-out) + D1 (warm) + D0 (cold) + D3 (ref) → E1 prewarmed shared memory → E2 (frozen artifact) → E5 → E6 → venue → fills → E7 → D4 async → nightly D6 research → E4 ladder → owner → D5 signed → Mode T boots it.

7-B Research → live ladder (the only promotion path)

COLLECT (R1-stamped) → LABEL (triple-barrier, net-of-costs, from signal log not fills) → SEARCH in Mode R (GP/ML/meta-label, **10^3-candidate budget per style**) → GATE E4 (CPCV purge+embargo by day, DSR ≥ 0.95 vs decorrelated trial count, crisis replay, net-of-cost) → PAPER (E6-PAPER) → SHADOW (3–6 mo) → owner approval (≤5/week) → D5 signed write → Mode T loads at next boot → MONITOR (rolling IC/PSI-drift/edge decay) → retrain loop or retire.

7-C Fix-a-bug loop (build/review/audit)

1. ticket from alert/chat (fleet-control seat repro'd via E7/D4 flight data) →
2. build seat: reproduce + fix + regression test (Rust: `cargo clippy -- -D warnings`
   - `cargo test`; Python: `uv run pytest`; TS seats: `tsc --noEmit` + test) →
3. review seat + cybersec seat (egress diff/marketing-drift/secret scan) →

4. owner merge → CI gates rerun → deploy window (no live-position freeze violation). **Cooldown rule:** cells touched by a fix restart in shadow first if E4 still binding.

7-D Training / self-learning / cross-cell merge (offline only)

- Weekly: re-run §7-B labels → candidate artifacts; every attempt G4-logged.
- Drift-triggered: edge decay (rolling net-IC vs frozen) → retrain+re-gate, logged.
- **Cross-cell merge:** monthly research batch flags overlapping artifacts (same signal family across cells) → merge candidate must re-clear **full E4** on the union instrument list, then the old artifacts retire **per-cell**, owner-approved.
- Hard ceiling unchanged: no live rewrite of Mode T; no LLM in R1/R2; no style enable without E4; agents never edit risk limits (RL-gate R7 of FULL-SYSTEM-PLAN).

7-E Cockpit chat loop (owner ⇄ system)

report types: cell health · portfolio · kill-state per market · pending L0–L3 approvals · sprint status · LLM-layer quality (scores passed E2 gate, token burn) · budget burn · ticket burn-down · research batch digest. Every chat line labeled ESTIMATE/MEASURED/SOURCE. Weekly digest auto-writes to the vault. Incidents page (owner mobile kill is out-of-band and always works without the app).

§8 · Comms & wiring (latency-engineering targets — ESTIMATE until measured at gates)

Hop	Transport	Budget
venue feed → R1 → D2 fan-out to cells	NATS JetStream WS	≤ ~2 ms internal P99 (MEASURE at W0b)
D1 warm read (boot/replay/dashboards)	QuestDB ILP	ms-class; not on tick path
LLM seat → task queue → seat	Postgres `llm_tasks` + NATS subject	seconds — intentionally async
cockpit chat → owner	gateway or thin WS (O16 decision)	human-scale
order path E1→E5→E6→wallet	in-process Rust + pre-serialized templates	<5 ms internal; venue RTT dominates (HFT honesty R8)
research scans	D6 DuckDB vectorized	seconds–minutes, off-peak

Container layer (owner ask 2, adopted): Compose wraps Postgres/QuestDB/NATS/ cockpit/chat + build workers; per-service CPU pinning (council fix A2) · resource volumes sized per store · restore drills reuse Compose snapshots · the 10×-restart W0a gate is defined against the Compose stack; **NOT** in the tick path (rail stays bare-metal).

§9 · Memory — every kind, every agent (owner ask 2/4)

Memory type	Non-LLM fleet	LLM seats	Store/writer
Working	cell process state (E1/E2/E5)	task-lease payload in `llm_tasks`	process / Postgres
Episodic	E7 ledger + G4 trial log	D4 audit of every run (token cost, result)	append-only
Semantic	R3 frozen artifacts + calendars/limits	D5 knowledge artifacts + Graphify graph.json + vault notes	signed/versioned
Procedural	code + runbooks (§6A)	skill files (§7 of LLM blueprint)	git + reviews
Owner rules	—	vault note (rules people + seat rules)	cockpit writes only

New layers (owner S44 ask, adopted in LLM-FLEET-BLUEPRINT §8.1, adopted again here):

- **Obsidian layer** = declarative decisions/knowledge: `gfi/` IS the vault already; add note types (decision · incident · runbook · research-digest · training-record)
  - cross-links; cockpit weekly digest auto-writes; read-only from agents.
- **Graphify layer** = code+docs knowledge graph on-demand: adopt as **local tool** (tree-sitter AST, 0-token code mode, EXTRACTED/INFERRED edge audit) refreshed via `--update` in the build/review loop; optional vault-index expansion — **not** the paid Cloud. Token-savings claim = author-reported/unreprovable → **W0c fit-test** measures our own delta before any agent depends on it.
- Recall substrate candidates (brigade Tideline patterns vs ruflo `@claude-flow/memory`) decided by a W0c toy-shootout (measured recall/latency), then wired to R3/D5 tables.

§10 · Quality machinery (the honest wall)

Gate	Threshold	Layer
DSR at E4	≥ 0.95 vs decorrelated trial count (G4)	research
Pressed edge-lift (E2)	beat frozen baseline in replay repeatedly	scoring seats
Crisis replay	EOD index-option crisis series pass	E4
Backtest discipline	CPCV purge + embargo, day-grouped folds	E4
Self-learn cap	10^3 candidates/style; if it can't clear DSR, no trade	G4/C3
Code	`cargo clippy -D warnings` / `cargo test` · `tsc --noEmit` · `uv run pytest` · review+cybersec seats · secret scan	all seats
LLM honesty	citations required (quant-mind contract); ESTIMATE/MEASURED labels in cockpit; anti-fabrication hard rules lifted from Vibe-Trading; **S44-gap-register G5** eval-replay harness with FunctionModel-style stubs	seats
Liquidity/cost	venue cost model in E5 governor + E7 reconciler	always

§11 · Build instructions — wave by wave, gate by gate

The 10-stage order (FULL-SYSTEM-PLAN §5) + S41/S42 amendments, now with **per-stage validation pins** (a wave is DONE only when its pins are true commands with true outputs):

Wave	Builds (from all 4 plans)	Validation pins
W0a	D0 lake · Compose stack (D3/D4/D5/D6 + NATS + QuestDB) · R1 rail skeleton · PIT replay harness · 5 D1 gates (burst/determinism/pre-warm/restore/10×-restart)	replay twice identical (bit-level) · harness outputs filed · result JSON in D3
B1 (S42)	collecting engines for market/news/world · backfill 6–10y, all 3 domains, TODO-driven · in-DB gaps table · blocked-source handling	domain verified + gaps table non-empty-but-justified → B-GATE before any training
W0b	D4 ledger + append-only roles · D5 signed registry · E1/E2/E5/E7 skeletons · L0–L4 cascade + out-of-band kill · IPC fan-out benchmark	crash-restart cycle ×10 · fan-out P99 file · kill drill rehearsal log
W0c (S44/45)	LLM control plane: brigade-guard port + ruflo swarm runner · `llm_tasks` · L0–L3 bridge · Graphify fold-in + Obsidian note types · G1 vault · G2/G9 dashboards · fit-test O16	toy build measures: cold start · 10×-restart of Compose · Graphify recall delta · substrate shootout
W1	D6 research cache · PySR/MAPIE arch integration · E4-SIM/BT/WF · E6-PAPER	one full style through §7-B to D5 (dry-run)
W2	seats live (news/data-QA/compliance) · cost model verified O8 ticking · proton JetStream-MV trial (gates: ingest parity ≥ JetStream benchmark) · G4/G6/G7	first paper fills · reconcile E7 vs broker
W3	first cells live-candidate · cTrader adapter (NY4 lane planning) · shadow 3–6 mo	shadow METRICS, no live claims
W4–W5	tick+/depth feeds mode · Hyperliquid node option · crypto cells · training/tech-selflearn rolling	drift-trigger retrain demonstrated
Stage 6+	cockpit GA (OpenStock-pattern UX ideas; O16 resolved)	L3 flows + weekly digest flowing
Stages 7–10	Terminal/Portal/Media (public, rights-firewalled) · World Monitor LAST · HFT subsystem only if measured demand	rights firewall live · Row-1 replica never blocks hot path

§12 · Monitor/control plan (owner ask 4 — the one line per surface)

1. **Cells**: NATS subjects + D3 cell-registry; per-cell heartbeat; L0–L4 map in cockpit.
2. **Engines**: p95/p99 dashboards (Rust metrics; Grafana-lite on Postgres, G2).
3. **Data**: gap calendar (B1 gate tables), D1/D0 budget burn e.g., 16-TB tracker.
4. **LLM layer**: token burn per seat/day (G9), truth labels, eval-harness replays.
5. **Owner mobile**: out-of-band kill + status, independent of everything else.

§13 · Honest risk register (standings, not new)

single-box fragility → offsite/backup story + compose snapshots; venue RTT domination (HFT honesty R8: no shortcut); vendor license drift (re-verify cadence); India data gaps pre-2022 = paid-vendor question (O12/O15); council single-source research claims (meta-labeling etc. —

C14: adopt cautiously, verify on our data); cockpit/LLM layer is NEW code with new failure modes — treat at parity with trading code for review gates.

§14 · Open decisions (O-registry status)

ID	Item	Needed by
—	~~O1~~ RESOLVED (S43)	—
O3	F&O cell granularity (per underlying vs per contract)	Stage-5 kickoff
O4	keep 7 off-index names or swap to true NIFTY 50	B1
O5	first style per instrument (73→75 brain count note)	build
O6	add FX scalping style? (30→31 styles)	build
O7	hedge-executor shape (per-cell vs portfolio)	build
O8	cTrader production limits (500/min is demo)	B1/W2
O9	Upstox algo-tagging 13th digit	W2
O10	order-balance requirement definition	B1
O11	Hyperliquid FIU-IND/FEMA/tax	W2 (before HL cells)
O12	PIT as-reported fundamentals source	B1
O13	ToS/ban-avoidance checklist	B1 sourcing
O14	model routing + privacy per LLM seat	W0c
O15	NSE index membership / survivorship-free vendor	W2
O16	cockpit runtime: brigade gateway vs thin own	W0c

§15 · Sources

Primary: the four specialist docs (each with its own sources S1–S17) + planning log S1–S44. New lines this session: consolidation synthesis over those docs; O-registry restated from log; no new external claims. External numbers cited earlier (71.5×, "80% cut", vendor benchmarks) remain labeled as before — **this doc adds no new uncorrected claims**.

Session 45 log entry records this deliverable; commit hash recorded in the log.